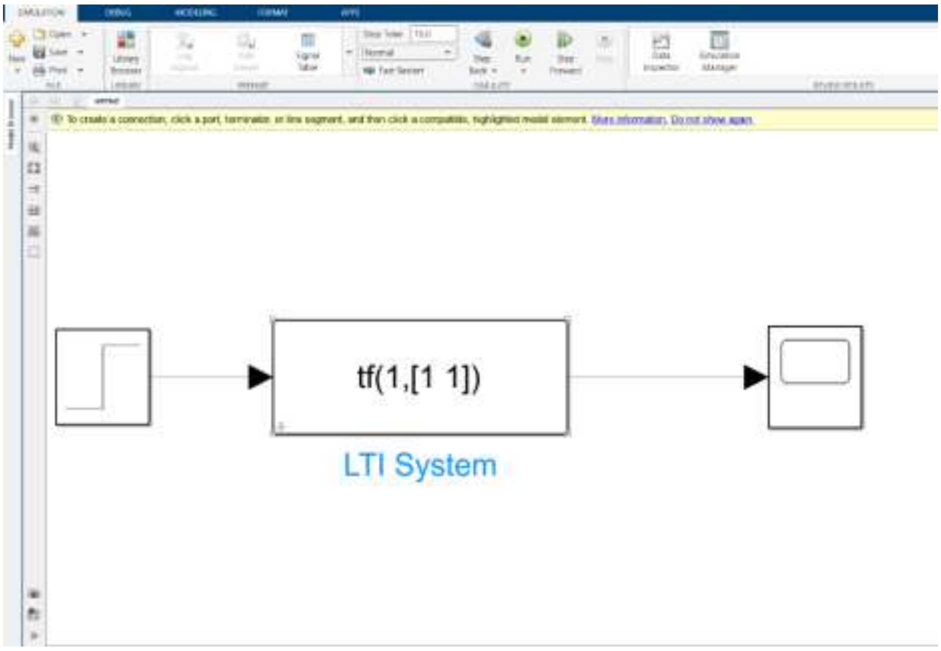
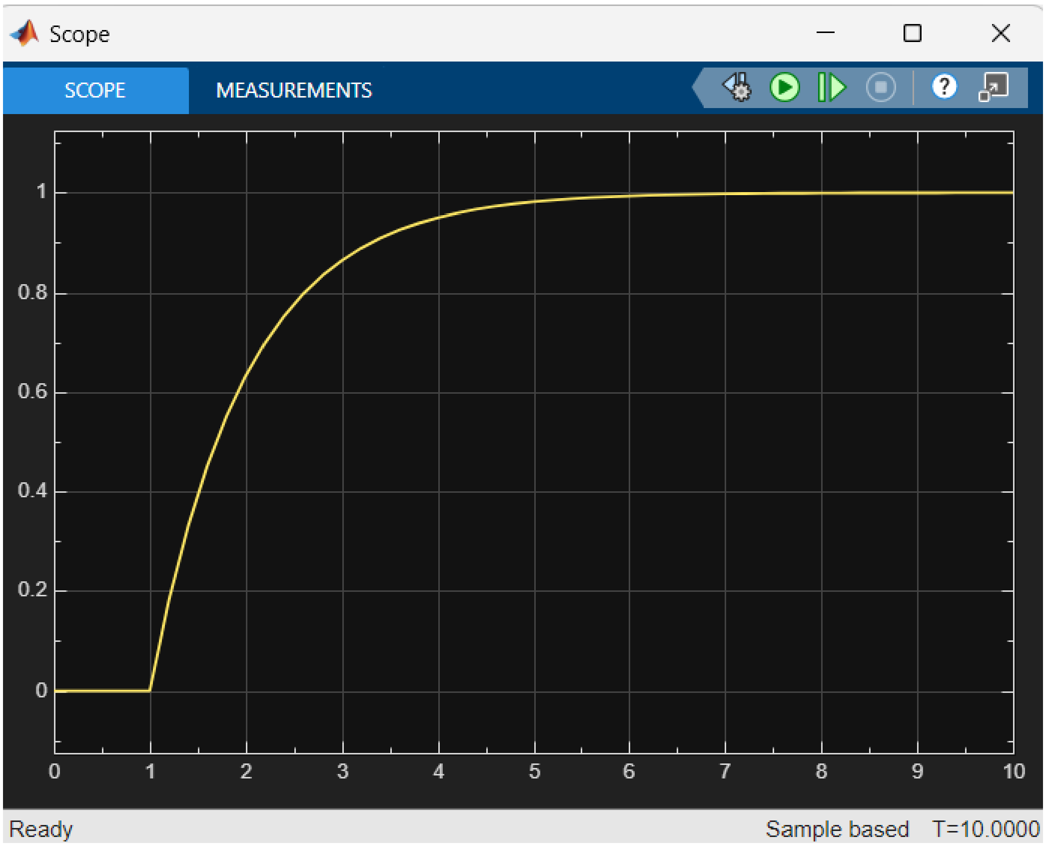
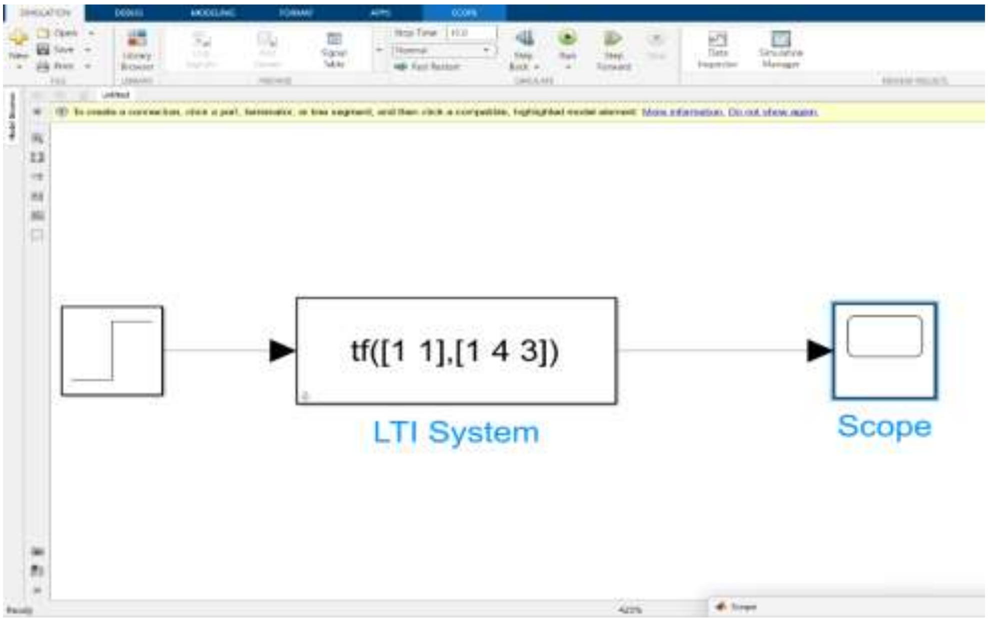
- **Plotting poles and zeros using Simulink:**
  - Diagram :



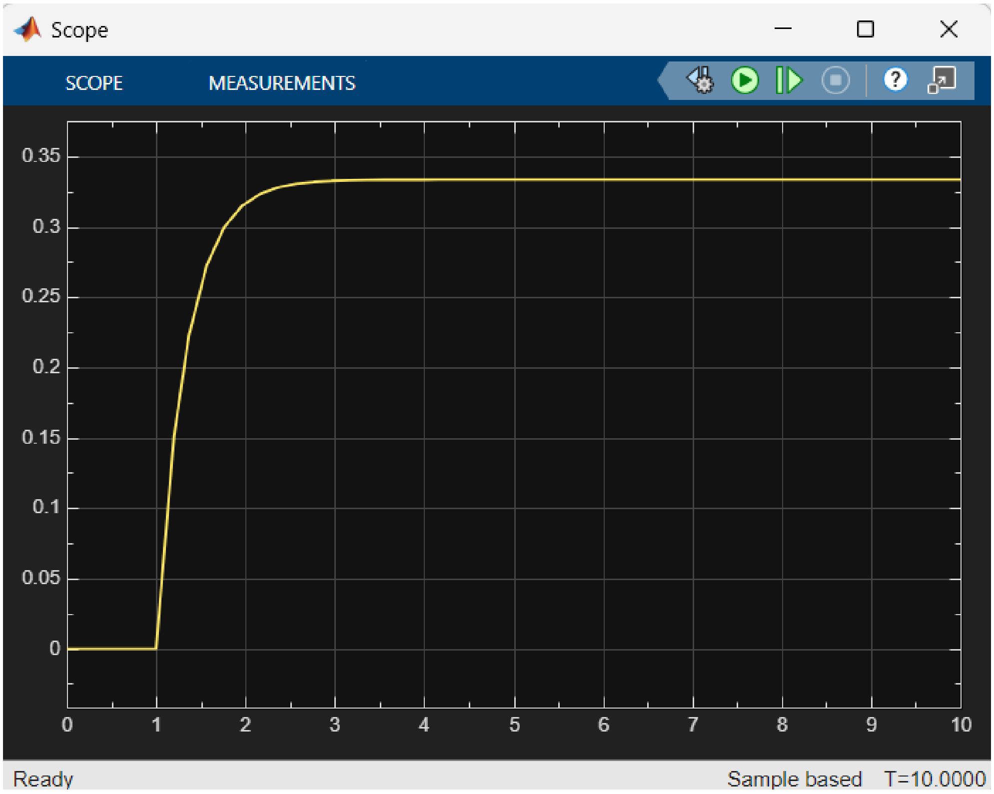
  - Output :



  - Circuit Diagram :



- Output :



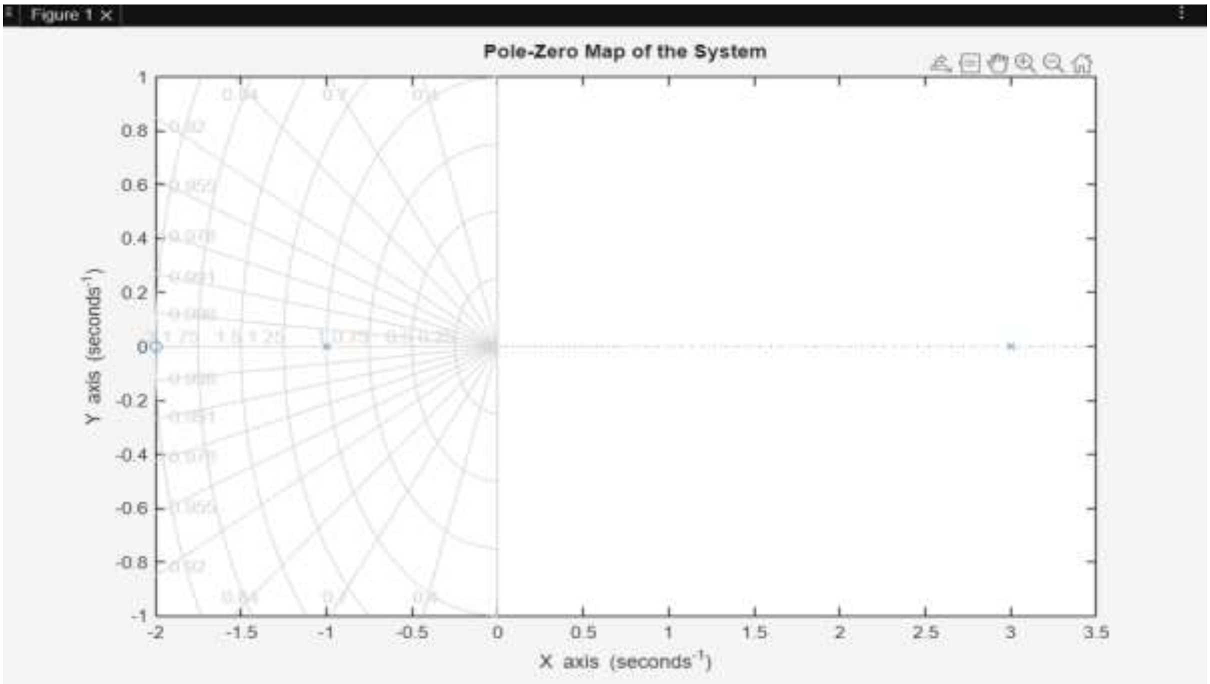
- **Plotting poles and zeros using code:**

```
numerator = [1 2];
denominator = [1 -2 -3];

system = tf(numerator, denominator);

pzmap(system);
grid on;
xlabel("X axis");
ylabel("Y axis");
title('Pole-Zero Map of the System');

poles = pole(system);
zeros = zero(system);

disp("The poles of the system: ");
disp(poles);

disp("The zeros of the system: ");
disp(zeros);

if all(real(poles) < 0)
    disp("The system is stable");
elseif any(real(poles) > 0)
    disp("The system is unstable");
else
    disp("The system is marginally stable");
end
```

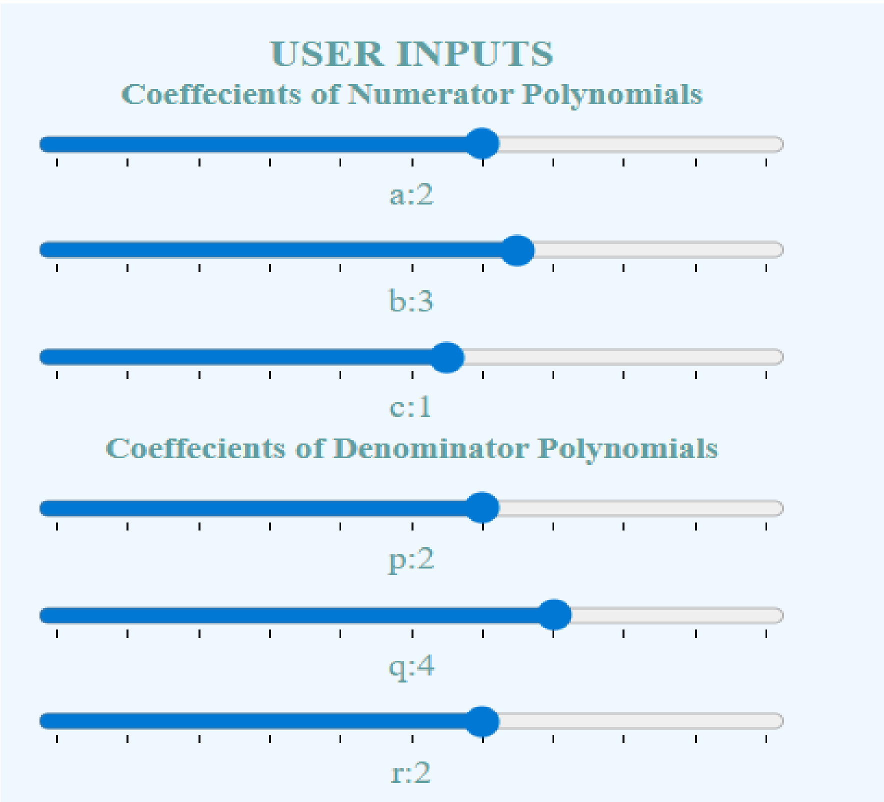
- Output :

```
The poles of the system:
    3.0000
   -1.0000

The zeros of the system:
    -2

The system is unstable
```

- Output :



- **Plotting poles and zeros of transfer function using Virtual lab:**

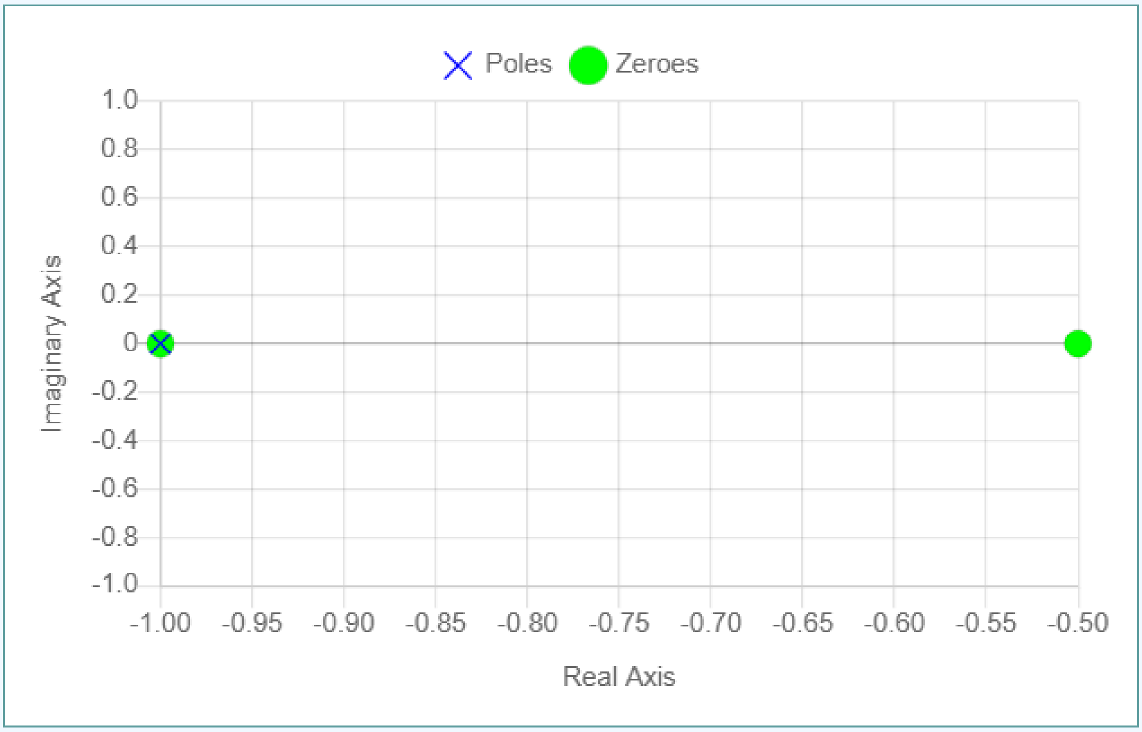


```
Number of zeroes

    2

Number of poles

    1

Generate transfer function
g =
```

$$\frac{2s^2+3s+1}{2s^2+4s+2}$$

```
Obtain pole,zero and gain values
Z(zeroes) =

   -1

   -0.5

P(poles) =

   -1

K(Gain) =

 1.00

plot values on graph
```

Results



- Conclusion :

As one pole of the system lies in left half of the s-plane, therefore the System is Stable